

Expert evaluation of BCHAIN-ACCESS

Expert panel: N = 10 multidisciplinary experts (≥ 5 years) — HCI/UX design, software engineering, healthcare informatics, assistive technology / occupational therapy, and blockchain systems.

1 · Heuristic evaluation — per-heuristic severity

Final severity per Nielsen heuristic (scale 0–4)



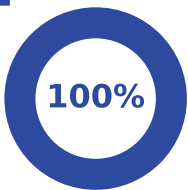
Mean 0.1 · max = 1 (H9) · 0 ratings ≥ 2 · N = 10 experts

Findings:

only H9 (Error Recovery) rated 1 (cosmetic); 0 serious or catastrophic issues.

Outcomemean severity 0.1 — below the 0.5 compliance gate.

2 · Cognitive walkthrough — task-based evaluation



8 tasks completed

0 errors

1.9 mean difficulty (/5)

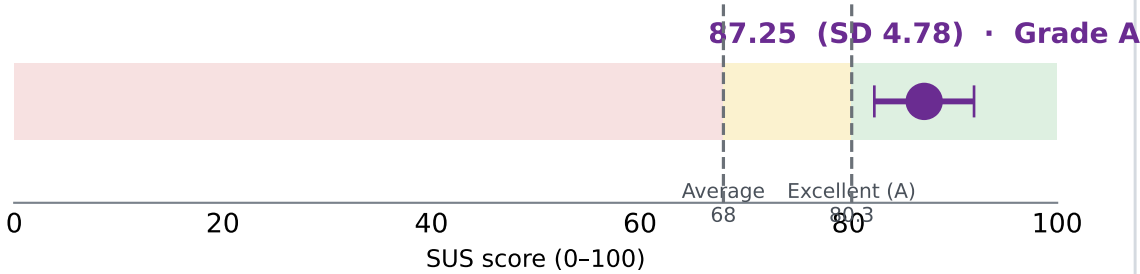
Tasks (T1–T8): Unlock · View Record · Check Access · Grant Access · Review · Confirm · Revoke · FAQ

Findings:

all 8 tasks completed without assistance; fixed countdown flagged for adjustability (E7, E9).

Outcome100% task success (80 / 80).

3 · System Usability Scale (Brooke, 1996)

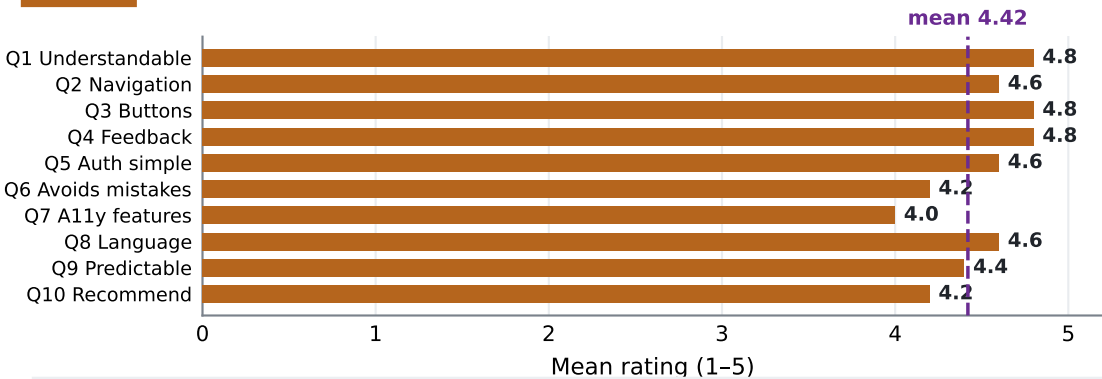


Findings:

87.25 exceeds the 80.3 'excellent (A)' threshold — far above the 68 average.

Outcomeusers rated the system as excellent.

4 · Accessibility questionnaire — item means (1–5)



Findings:

all 10 items ≥ 4.0 (high 4.8, low 4.0).

Outcomehigh accessibility satisfaction — overall 4.42 / 5.

Overall outcome — converging evidence across all four methods indicates BCHAIN-ACCESS is usable, understandable, and accessible, supporting inclusive interaction while maintaining security and compliance.

Next step — empirical validation with people with disabilities (PwDs) in real-world healthcare environments.